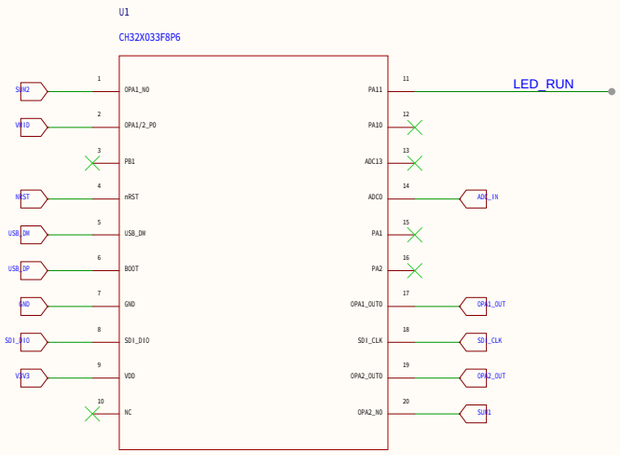
Aster-MCA-Nano-CH32-v0.3-USB-A-LED

01 MCU and analog pulse input | ENGINEERING PROTOTYPE - performance unverified

SMA -> AC coupling -> OPA2 LPF -> OPA1 LPF -> PA0 / ADC0. Both OPA + inputs = pin 2 / VMID.

External feedback. R9 fitted: gain ~0.199. R9 removed: gain ~1. Bandwidth and resolution require bench tests.



J2
132289 SMA-F



R5
1M



C4
33n C0G 50V



R6
10k 1%



C5
470p C0G 50V



R7
100k 0.1%



R8
110k 0.1%



R9
27.4k 0.1%



D2
BAT54S



R10
10k 1%



C6
470p C0G 50V



R11
100k 0.1%



R12
110k 0.1%



R13
10k 0.1%



R14
10k 0.1%



C7
4.7u X7R 10V



C8
100n X7R 10V



R15
100R 1%



C9
1n C0G 50V



TP6
VMID



TP7
OPA2_OUT



TP8
ADC_IN



Aster-MCA-Nano-CH32-v0.3-USB-A-LED

02 USB-A, power and boot | ENGINEERING PROTOTYPE - performance unverified

USB ISP: hold BOOT while plugging in USB, release, program using WCHISPTool. Preserve factory bootloader.

D1
USBLC6-2SC6



U2
XC6206P332MR-G



C1
1u X7R 10V



C2
4.7u X7R 10V



C3
100n X7R 10V



SW1
B3U-1000P BOOT



R16
1k



R17
10k



TP1
GND



TP2
V3V3



TP3
SDI_DIO



TP4
SDI_CLK



TP5
NRST

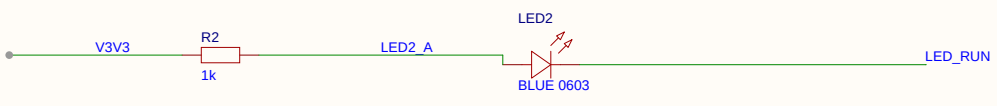


J1
1001-011-01101





PWR: 3.3V rail



RUN: active low, 1Hz heartbeat